

RF-65-NUTRIENTS

1.SMOTE

Average Accuracy Val K-Fold: 91.6661

Average Accuracy Train K-Fold: 99.8926

Average MCC Val K-Fold: 0.8196

Average MCC Train K-Fold: 0.9986

Average F1 Score Val K-Fold: 0.9173

Average F1 Score Train K-Fold: 0.9989

Average Precision Val K-Fold: 0.9192

Average Precision Train K-Fold: 0.9989

Average Recall Val K-Fold: 0.9167

Average Recall Train K-Fold: 0.9989

Accuracy Test: 89.06%

MCC Test: 0.7631

F1 Score Test: 0.8912

Precision Test: 0.8922

Recall Test: 0.8906

Classification Report:

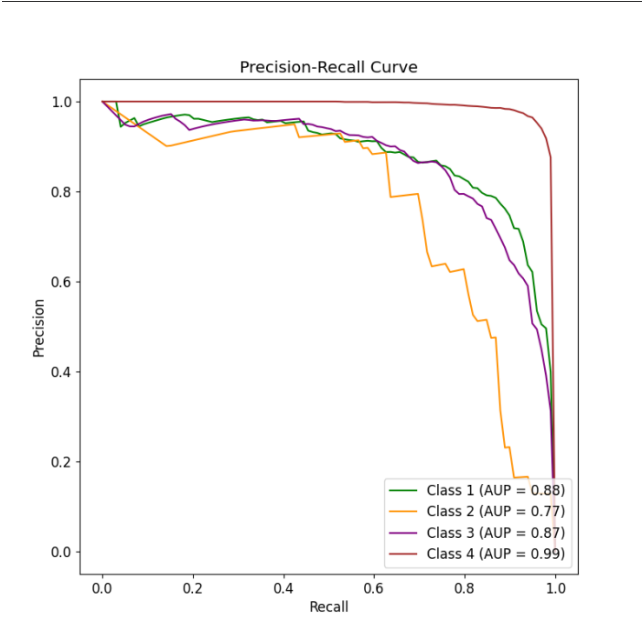
	precision	recall	f1-score	support
1	0.74	0.76	0.75	68
2	0.69	0.82	0.75	11
3	0.73	0.73	0.73	93
4	0.96	0.95	0.95	422
accuracy			0.89	594
macro avg	0.78	0.82	0.80	594
weighted avg	0.89	0.89	0.89	594

Confusion Matrix Test:

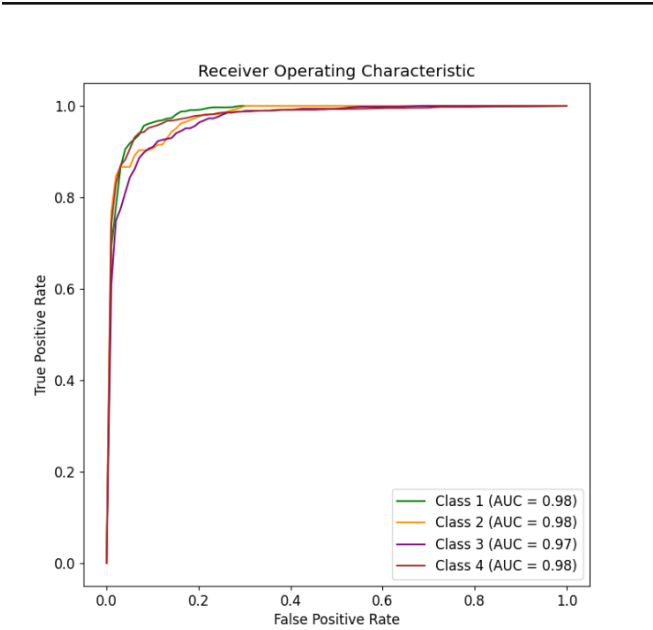
```
[[ 52  0 12  4]
```

[0 9 0 2]
[12 1 68 12]
[6 3 13 400]]

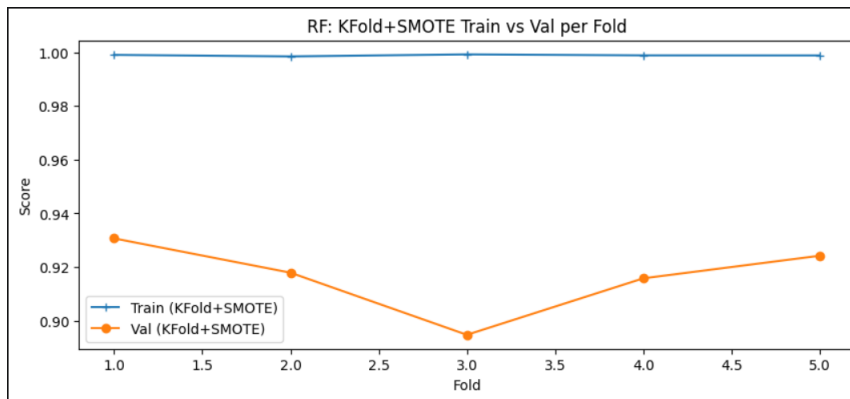
PR curve



ROC curve



Training and Validation curve



2.SMOTE+STRATIFIED

Average Accuracy Val K-Fold: 91.2459

Average Accuracy Train K-Fold: 99.8928

Average MCC Val K-Fold: 0.8116

Average MCC Train K-Fold: 0.9986

Average F1 Score Val K-Fold: 0.9124

Average F1 Score Train K-Fold: 0.9989

Average Precision Val K-Fold: 0.9153

Average Precision Train K-Fold: 0.9989

Average Recall Val K-Fold: 0.9125

Average Recall Train K-Fold: 0.9989

Accuracy Test: 90.24%

MCC Test: 0.7839

F1 Score Test: 0.9012

Precision Test: 0.9010

Recall Test: 0.9024

Classification Report:

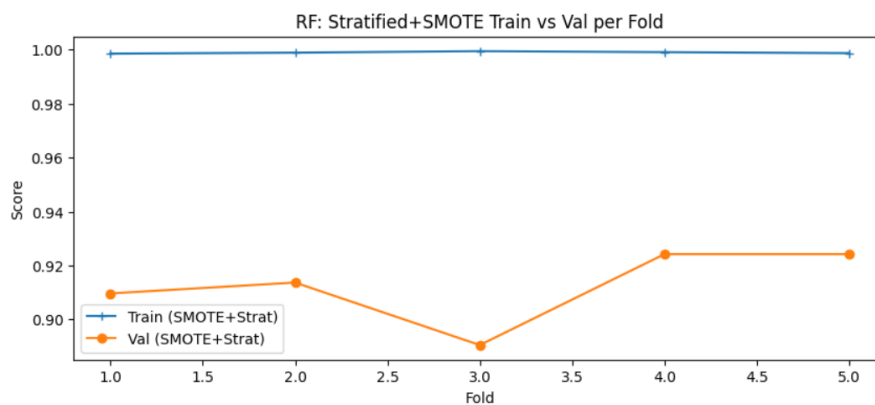
	precision	recall	f1-score	support
1	0.75	0.79	0.77	68
2	0.90	0.82	0.86	11
3	0.80	0.71	0.75	93
4	0.95	0.96	0.96	422

accuracy		0.90	594
macro avg	0.85	0.82	0.83
weighted avg	0.90	0.90	0.90

Confusion Matrix Test:

```
[[ 54  0  9  5]
 [  0  9  0  2]
 [ 12  0 66 15]
 [  6  1  8 40]]
```

Training and Validation curve



3.STRATIFIED

Average Accuracy Val K-Fold: 90.7830

Average Accuracy Train K-Fold: 99.6949

Average MCC Val K-Fold: 0.7913

Average MCC Train K-Fold: 0.9933

Average F1 Score Val K-Fold: 0.9044

Average F1 Score Train K-Fold: 0.9969

Average Precision Val K-Fold: 0.9064

Average Precision Train K-Fold: 0.9970

Average Recall Val K-Fold: 0.9078

Average Recall Train K-Fold: 0.9969

Accuracy Test: 90.07%

MCC Test: 0.7754

F1 Score Test: 0.8969

Precision Test: 0.8961

Recall Test: 0.9007

Classification Report:

	precision	recall	f1-score	support
1	0.81	0.75	0.78	68
2	0.90	0.82	0.86	11
3	0.79	0.67	0.73	93
4	0.93	0.98	0.95	422
accuracy		0.90		594
macro avg	0.86	0.80	0.83	594
weighted avg	0.90	0.90	0.90	594

Confusion Matrix Test:

[[51 0 11 6]

[0 9 0 2]

[9 0 62 22]

[3 1 5 413]]

Training and Validation curve

